Lagrange Interpolation Polynomial

$f(x) = L_1 f(x_1) + L_2 f(x_2)$

$L_1 = \frac{x - x_2}{x_1 - x_2} : \begin{cases} 1, & x = x_1 \\ 0, & x = x_2 \end{cases}$

$L_2 = \frac{x - x_1}{x_2 - x_1} : \begin{cases} 0 & x_1 \\ 1 & x_2 \end{cases}$

$f(x) = \frac{x - x_2}{x_1 - x_2} f(x_1) + \frac{x - x_1}{x_2 - x_1} f(x_2)$

$f_{n-1}(x) = \sum_{i=1}^{n} L_i(x) f(x_i)$

$L_i(x) = \prod_{\substack{j=1 \\ j \neq i}}^{n} \frac{x - x_j}{x_i - x_j}$

$f(x_3) \cdot L_3 = \frac{(x - x_1)(x - x_2)}{(x_3 - x_1)(x_3 - x_2)} f(x_3)$

2nd order

$f_2(x) = \frac{(x - x_2)(x - x_3)}{(x_1 - x_2)(x_1 - x_3)} f(x_1) + \frac{(x - x_1)(x - x_3)}{(x_2 - x_1)(x_2 - x_3)} f(x_2) + \frac{(x - x_1)(x - x_2)}{(x_3 - x_1)(x_3 - x_2)} f(x_3)$

Extrapolation

Process of estimating $f(x)$ at an x value outside the range of known base pts: $x_1, x_2, \dots x_n$

Extrapolation must be done with extreme case

oscillations: high-order polynomials cause oscillation very sensitive to rounding error

Splines and Piecewise Interpolation

use ~~to~~ lower-order polynomials in a piecewise way, using splines

Linear Splines

$S_i(x) = a_i + b_i (x - x_i)$ $a_i = f_i$ $b_i = \frac{f_{i+1} - f_i}{x_{i+1} - x_i}$

left pt: (x_i, f_i)

right pt: (x_{i+1}, f_{i+1}) newton's 1st polyn

$S_i(x) = f_i + \frac{f_{i+1} - f_i}{x_{i+1} - x_i} (x - x_i)$

$S_i(x) = a_i + b_i (x - x_i) + c_i (x - x_i)^2$

$S_i(x) = a_i + b_i (x - x_i) + c_i (x - x_i)^2 + d_i (x - x_i)^3$

Parabolic Interpolation

Initial guesses

Fit a parabola to the pts $(x_1, f(x_1)), (x_2, f(x_2))$ $(x_3, f(x_3))$ differentiate it and set the result equal to zero and solve for an estimate of the optimal x

$$x_4 = x_2 - \frac{1}{2} \frac{(x_2 - x_1)^2 [f(x_2) - f(x_3)] - (x_2 - x_3)^2 [f(x_2) - f(x_1)]}{(x_2 - x_1)[f(x_2) - f(x_3)] - (x_2 - x_3)[f(x_2) - f(x_1)]}$$

Piecewise

Spline: cubic spline interpolation

pchip: piecewise cubic hermite

interp1: Spline and Hermite

yy = spline(x, y, xx) Interp(x, y, x_i, 'method').

nearest: nearest neighbor interpolation (0 order)

linear: Straight lines to connect the pts

pchip & cubic: piecewise cubic Hermite interpolation

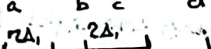
Cubic Hermite:

- continous first derivatives
- second derivatives are not necessarily continous
- interpolation is shape preserving
- interpolated values do not tend to overshoot the data points as can sometimes happen in cubic splines

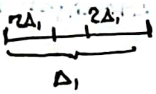
Optimization

Optimization deals with finding the minima and maxima of function depends on one or more variables. The goal is to determine the values of the variables that give the minima or maxima of the function

Golden Section Method

Iter 1: 

evaluate two intermediate pts. and get an idea of slope

iter 2: 

iter 3: 

If $f(b) < f(c)$ choose ac else bd

$\tau = \frac{3 - \sqrt{5}}{2} = 0.3820$

Golden-ratio: 

$\frac{l_1 + l_2}{l_1} = \frac{l_1}{l_2}$ multiply by $\frac{l_1}{l_2}$ and collect terms

$\frac{l_1}{l_2} + 1 = \frac{l_1^2}{l_2^2}$ $\phi = \frac{1 + \sqrt{5}}{2} = 1.618$

$\phi - 1$ = golden ratio

Golden Section Search

determine intermediate pts x_1, x_2

$x_1 = x_e + d$ $x_2 = x_u - d$ $d = \underbrace{(\phi - 1)}_{0.618 \approx 0.62} (x_u - x_e)$

i) choose two interior pts ii) define a new interval

ii) $f(x_1) < f(x_2)$ then $f(x_1)$ is the min, eliminate region x_e to x_2 replace x_e with x_2 old x_1 becomes x_2 $x_1 = x_e + d_{new}$

$f(x_2) < f(x_1)$ then $f(x_2)$ is the min, eliminate x_1 to x_u replace $x_1 = x_2$ x_2 becomes x_1, $x_2 = x_u - d_{new}$ $\varepsilon_a = (2 - \phi) \left| \frac{x_u - x_e}{x_{opt}} \right| \times 100\%$

Find the minimum $f(x) = \frac{x^2}{10} - 2 \sin x$ $x_e = 0$ to $x_u = 4$

$d = 0.618(4 - 0) = 2.4721$, $f(x_1) = -0.63$ $f(x_2) = -1.7647$

$x_1 = x_e + d = 2.4721$ $x_2 = x_u - d = 1.5279$ $f(x_2) < f(x_1)$ best estimate of min is at $x_2 = 1.5279$

min is in the interval x_e, x_2 and x_1 $x_e = 0$ $x_u = x_1 = 2.4721$ $x_1 = x_2 = 1.5279$

$f(x_1) = -1.7647$ $d_{new} = .62(2.4721 - 0) = 1.5279$ $x_2 = x_u - d = 2.4721 - 1.5279 = .944$

$f(x_2) = -1.5310$ iteration 2) $f(x_2) > f(x_1)$ → best min at $x_1 = 1.5279$ in interval (x_2, x_1, x_u)

[xmin, fval] = fminbnd(func, x1, x2)

mean(s) median(s) std(s) ... hist(s) + 10 bins hist(s, n)

rand(m, n) → random numbers btwn 0-1

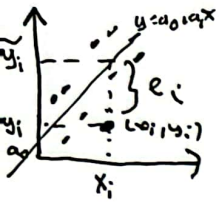
↳ linear down + (up - low) * rand(m, n)

Newton's Method

$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$ to find optimum of $f(x)$ $g(x) = f'(x)$ $f'(x^*) = g'(x^*) = 0$ $x_{i+1} = x_i - \frac{f'(x_i)}{f''(x_i)}$

Linear Regression

Std dev. $s_y^2 = \frac{\sum (y_i - \bar{y})^2}{n-1}$ coefficient variation $= \frac{s_y}{\bar{y}} \times 100\%$ Normal dist. $N(\mu, \sigma)$ ↳ Linear ↳ Std



$(x_1, y_1), (x_2, y_2) \dots (x_n, y_n)$ straight line $\tilde{y}_i = \overbrace{a_0 + a_1 x_i}^{\text{params}} + e_i$ ↳ residual btwn model and observations

$e_i = y_i - a_0 - a_1 x_i$ Criteria for best fit $\sum_{i=1}^{n} e_i = \sum_{i=1}^{n} (y_i - a_0 - a_1 x_i)$ not suitable since ↳ also absval.

fminsearch(fun, x_0, opts, p1, p2..) = [x, fval] polyfit(x, y, n) ↳ Lindeman

Sum of Least Squares

$S_r = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - a_0 - a_1 x_i)^2$ least squares gives a unique line for given set of data

$\frac{\partial S_r}{\partial a_0} = -2 \sum_{i=1}^{n} (y_i - a_0 - a_1 x_i) = 0$

$\frac{\partial S_r}{\partial a_1} = -2 \sum_{i=1}^{n} (y_i - a_0 - a_1 x_i) x_i = 0$

$\sum y_i - \sum a_0 - \sum_{i=1}^{n} a_1 x_i = 0$

$\sum x_i y_i - \sum a_0 x_i - \sum a_1 x_i^2 = 0$ $a_1 = \frac{n \sum x_i y_i - (\sum x_i)(\sum y_i)}{n \sum x_i^2 - (\sum x_i)^2}$ $a_0 = \bar{y} - a_1 \bar{x}$

Quantification of error of linear regression

$S_r = \sum_{i=1}^{n} (y_i - a_0 - a_1 x_i)^2$ $S_t = \sum (y_i - \bar{y})^2$ Standard error of estimate $= \sqrt{\frac{S_r}{n-2}}$ correlation coef. $r^2 = \frac{S_t - S_r}{S_t}$

$S_r = S_t$ $r^2 = 0$ the line fit gives no improvement $0 \le r \le 1$

$S_t - S_r$ = improvement or error reduction due to using straight line rather than $\bar{y}$

$S_r = 0$. $r^2 = 1$ → perfect fit

Linearization: $y = \alpha_1 e^{\beta_1 x} \xrightarrow{\ln} \ln y = \ln \alpha_1 + \beta_1 x$ $y = \alpha_2 x^{\beta_2} \xrightarrow{\log} \log y = \log \alpha_2 + \beta_2 \log x$

$y = \alpha_3 \frac{x}{\beta_3 + x}$ (population growth model) $\rightarrow \frac{1}{y} = \frac{1}{\alpha_3 \frac{x}{\beta_3 + x}} = \frac{\beta_3 + x}{\alpha_3 x} = \frac{1}{\alpha_3} + \frac{\beta_3}{\alpha_3} \frac{1}{x}$

$y = \alpha_2 x^{\beta_2} \rightarrow \log y = \log \alpha_2 + \beta_2 \log x$ $x_i, y_i \rightarrow \log x_i, \log y_i$ $x_i^2 \rightarrow (\log x_i)^2$

General Linear Least-squares and Nonlinear Equation

$y = a_0 + a_1 x + a_2 x^2 + e$ $S_r = \sum_{i=1}^{n} (\overbrace{y_i - (a_0 + a_1 x_i + a_2 x_i^2)}^{\text{true} \quad \text{estimate}})^2$ / $\frac{\partial S_r}{\partial a_0} = 0 \rightarrow -2 \sum_i (y_i - a_0 - a_1 x_i - a_2 x_i^2) = 0$

$n a_0 + (\sum x_i) a_1 + (\sum x_i^2) a_2 = \sum y_i$

$(\sum x_i) a_0 + (\sum x_i^2) a_1 + (\sum x_i^3) a_2 = \sum x_i y_i$

$(\sum x_i^2) a_0 + (\sum x_i^3) a_1 + (\sum x_i^4) a_2 = \sum x_i^2 y_i$

$y = a_0 + a_1 x_1 + \dots + a_m x^m + e$

$\frac{\partial S_r}{\partial a_1} = 0 \rightarrow -2 \sum x_i (y_i - a_0 - a_1 x_i - a_2 x_i^2) = 0$

$\frac{\partial S_r}{\partial a_2} = 0 \rightarrow -2 \sum x_i^2 (y_i - a_0 - a_1 x_i - a_2 x_i^2) = 0$

Multiple Linear Regression

$y = a_0 + a_1 x_1 + a_2 x_2 + e$ $S_r = \sum_{i=1}^{n} (\overbrace{y_i - (a_0 + a_1 x_{1,i} + a_2 x_{2,i})}^{\text{true} \quad \text{estimate}})^2$ → regression plane

$y = a_0 + a_1 x_1 + a_2 x_2 + \dots + a_m x_m + e$

$$\begin{bmatrix} n & \sum x_{1,i} & \sum x_{2,i} \\ \sum x_{1,i} & \sum x_{1,i}^2 & \sum x_{1,i} x_{2,i} \\ \sum x_{2,i} & \sum x_{1,i} x_{2,i} & \sum x_{2,i}^2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_{1,i} y_i \\ \sum x_{2,i} y_i \end{bmatrix}$$

Power Eq: $y = a_0 x_1^{a_1} x_2^{a_2} \dots x_m^{a_m}$ $\log y = \log a_0 + a_1 \log x_1 + \dots + a_m \log x_m$

Generalized Linear Least Squares $y = a_0 z_0 + a_1 z_1 \dots + e$ z : basic function

multiple linear regression: $z_0 = 1, z_1 = x_1, z_2 = x_2 \dots z_m = x_m$ polynomial regression: $z_0 = 1, z_1 = x, z_2 = x^2 = x_m$

fourier analysis $y = a_0 + a_1 \cos(\omega x) + a_2 \sin(\omega x)$ truly nonlinear: $y = a_0 (1 - e^{-a_1 x})$

Nonlinear Regression

$y = a_0 (1 - e^{-a_1 x}) + e$ $f(a_0, a_1) = \sum_{i=1}^{n} [\overbrace{y_i - a_0 (1 - e^{-a_1 x_1})}^{e_i}]^2$

Richardson Extrapolation

$h_2 = h_1/2$ $\quad D = \frac{4}{3}D(h_2) - \frac{1}{3}D(h_1)$

Forward diff.

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i} + O(h)$$

Newton-Cotes

$$I = \int_a^b f(x)dx \approx \int_a^b f_n(x)\,dx$$

$f_n(x) = a_0 + a_1 x + \ldots + a_n x^n$

Trapezoidal

$$I = (b-a)\frac{f(a)+f(b)}{2} \quad ? \text{ (between } a \text{ and } b)$$

$$E_t = \frac{-1}{12} f''(\xi)(b-a)^3$$

Composite Trapezoidal

$$I = \underbrace{(b-a)}_{\text{width}} \cdot \underbrace{\frac{f(x_0) + 2\sum_{i=1}^{n-1} f(x_i) + f(x_n)}{2n}}_{\text{avg height}}$$

$$E_t = -\frac{(b-a)^3}{12n^3}\sum_{i=1}^{n} f''(\xi_i)$$

Simpson's Rule

(more accurate than trapz)

$$I = \int_{x_0}^{x_2} \left[\frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} f(x_0) + \underbrace{\frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} f(x_1) + \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} f(x_2)}_{2^{nd}\text{ order polynomial } f_2(x)} \right] dx$$

Three pts $(x_0, f(x_0))(x_1, f(x_1))(x_2, f(x_2))$

1/3 Rule

$$I = \frac{h}{3}\left[f(x_0) + 4f(x_1) + f(x_2)\right] \qquad h = (b-a)/2 \quad a = x_0 \quad b = x_2$$

$$I = \frac{(b-a)}{2h}\,\frac{f(x_0) + 4f(x_1) + f(x_2)}{6}$$

Composite Simpson's 1/3 Rule

$$I = (b-a)\frac{f(x_0) + 4\sum_{i=1,3,5}^{n-1} f(x_i) + 2\sum_{j=2,4,6}^{n-2} f(x_j) + f(x_n)}{3n}$$

Simpson's 3/8 Rule

$$I = \frac{3h}{8}\left[f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)\right]$$

$h = \frac{b-a}{3}$

MATLAB: trapz, cumtrapz

z = trapz(x, y)

Integration with Unequal Segment

$$I = h_1\frac{f(x_0)+f(x_1)}{2} + h_2\frac{f(x_1)+f(x_2)}{2} + \ldots + h_n\frac{f(x_{n-1})+f(x_n)}{2}$$

Multiple Integrals

$$I = \int_c^d \left[\int_a^b f(x,y)\,dx\right] dy = \int_a^b \left[\int_c^d f(x,y)\,dy\right] dx$$

$$\bar{f} = \frac{I}{(b-a)(d-c)}$$ avg value of $f(x,y)$

MATLAB: dblquad(fun, xmin, xmax, ymin, ymax, tol)

Matrices

Symmetric: $a_{ij} = a_{ji}$

Diagonal elements are zero

Banded matrix:

$$[A] = \begin{bmatrix} a_{11} & 0 & 0 \\ a_{21} & & \\ 0 & & \\ 0 & 0 & \end{bmatrix}$$ bandwidth 3, tridiagonal matrix

$[C] = [A][B]$, $c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$

Cramer's Rule

$Ax = b$ $\quad x_1 = \begin{vmatrix} b_1 & c_{21} & c_{31} \\ b_2 & c_{22} & c_{32} \\ b_3 & c_{23} & c_{33} \end{vmatrix} / D$

LU Factorization

$Ax = b \rightarrow Ax - b = 0$

$$\underbrace{\begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}}_{U} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}}_{d}, \quad L = \begin{bmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{bmatrix}$$

$Ld = b$

$Ux - d = 0$ $\quad L(Ux-d) = Ax - b$ $\quad LU = A$

$Ax = b$ → $Ux = d$, $Ld = b$ → solve for d → solve for x

$[L,U] = \text{lu}(A)$

$d = L \backslash b$

$x = U \backslash d$

Gauss Elimination as LU

$A = LU$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

multiply row 1 by $f_{21} = \frac{a_{21}}{a_{11}}$ subtract from row 2

multiply row 1 by $f_{31} = \frac{a_{31}}{a_{11}}$ " " row 3

multip. the modified row 2 by $f_{32} = \frac{a'_{32}}{a'_{22}}$ & from 3rd row

$$L = \begin{bmatrix} 1 & 0 & 0 \\ f_{21} & 1 & 0 \\ f_{31} & f_{32} & 1 \end{bmatrix} \qquad U = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & 0 & a'_{33} \end{bmatrix}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ f_{21}a_{11} & f_{21}a_{12} + a'_{22} & f_{21}a_{13} + a'_{23} \\ f_{31}a_{11} & f_{31}a_{12} + f_{32}a'_{22} & f_{31}a_{13} + f_{32}a'_{23} + a'_{33} \end{bmatrix}$$

$f_{21} = \frac{a_{21}}{a_{11}}$ $\quad f_{31} = \frac{a_{31}}{a_{11}}$ $\quad f_{32} = \frac{a'_{32}}{a'_{22}}$

Solve for d from $Ld = b$, x from $Ux = d$

$$7x_1 + 2x_2 - 3x_3 = -12$$
$$2x_1 + 5x_2 - 3x_3 = -20$$
$$x_1 - x_2 - 6x_3 = -26$$

$$A = \begin{bmatrix} 7 & 2 & -3 \\ 2 & 5 & -3 \\ 1 & -1 & -6 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ \frac{2}{7} & 1 & 0 \\ \frac{1}{7} & -0.429 & 1 \end{bmatrix}}_{L}\underbrace{\begin{bmatrix} 7 & 2 & -3 \\ 0 & 4.429 & -2.143 \\ 0 & 0 & -6.165 \end{bmatrix}}_{U}$$

Error Analysis and System Cond.

System = $Ax = b$

1. Scale A so that largest element in each row 1. Invert scaled A if some elements of A^{-1} are several order of magnitude greater than 1, likely that system ill-conditioned
2. Multiply the inverse by the original coefficient matrix and see whether the result is close to identity. If not ill-conditioned
3. Invert the inverted matrix and see if it is sufficiently close to the original matrix. If not ill-conditioned.

Vector and Matrix Norms

$x = [x_1, x_2 \ldots x_n]$ $\quad \|x\|_e = \sqrt{\sum_{i=1}^{n} x_i^2}$ euclidean norm

Matrices: frobenius norm: $\|A\| = \sqrt{\sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}^2}$ A single value that quantifies the "size" of A

p norms: $\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p}$ $\quad p=2$ euclidean norm $\quad p=1$ L_1 norm $\quad [L_2 \text{ norm}]$

$\|x\|_1 = \sum_{i=1}^{n} |x_i|$ $\quad \|x\|_\infty = \max|x_i|$ $\quad 1 \le i \le n$

Matrices: $\|A\|_1 = \max_{1\le j\le n} \sum_{i=1}^{n} |a_{ij}|$ column-sum norm

$\|A\|_\infty = \max_{1\le i\le n} \sum_{j=1}^{n} |a_{ij}|$ row-sum norm

spectral norm: $\|A\|_2 = (\mu_{max})^{1/2}$ Largest eigen value of $A^T A$

Matrix Condition Number

$\text{Cond}(A) = \|A\|\,\|A^{-1}\|$ ill-conditioned if $\text{cond}(A) \gg 1$

$$\begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{bmatrix}$$ 3×3 hilbert matrix

use row-sum norm to estimate the condition number

hilbert matrix is ill-conditioned

norm(X,p) cond(X,p)

$$A^{-1} = \begin{bmatrix} 9 & -18 & 10 \\ -36 & 96 & -60 \\ 30 & -90 & 60 \end{bmatrix} \rightarrow \|A^{-1}\|_\infty = |-36| + |96| + |-60| = 192$$

if the largest element of $A^{-1} \sim 1$ system can be considered well-conditioned

sum each row

$\|A\|_\infty = 2.35$

$\text{cond}(A) = (2.35)(192)$

$= 451.2$

Permutation Matrix

i) $PA = LU$ $\quad P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

ii) $PAx = LUx = Pb \rightarrow Ld = Pb$ Solve for d

iii) $Ux = d$ $\quad \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}\begin{bmatrix} 2 & -7 & 4 \\ 7 & 3 & 6 \\ 5 & 1 & 9 \end{bmatrix} = \begin{bmatrix} 5 & 1 & 9 \\ 7 & 3 & -6 \\ 2 & -7 & 4 \end{bmatrix}$

Determinant

$$D = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$$\tilde{D} = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{21}a_{12}$$

Gauss Siedel

$Ax=b$ (iterative)

$$x_1^j = \frac{b_1 - a_{12}x_2^{j-1} - a_{13}x_3^{j-1}}{a_{11}} \quad (1) \qquad x_2^j = \frac{b_2 - a_{21}x_1^j - a_{23}x_3^{j-1}}{a_{22}} \qquad x_3^j = \frac{b_{38} - a_{31}x_1^j - a_{32}x_2^j}{a_{33}}$$

Converge criterion: $\varepsilon_{a,i} = \left|\frac{x_i^j - x_i^{j-1}}{x_i^j}\right| \times 100\% \le \varepsilon_s$

Jacobi Iterations

$$x_1^j = \frac{b_1 - a_{12}x_2^{j-1} - a_{13}x_3^{j-1}}{a_{11}} \qquad x_2^j = \frac{b_2 - a_{21}x_1^{j-1} - a_{23}x_3^{j-1}}{a_{22}} \qquad x_3^j = \frac{b_3 - a_{31}x_1^{j-1} - a_{32}x_2^{j-1}}{a_{33}}$$

Gauss siedel will converge $|a_{ii}| > \sum_{j=1}^{n} |a_{ij}|$ } diagonally dominant system — sufficient but not neccessary condition

Successive-Substitution (= fixed-point iteration)

$x_1^2 + x_1x_2 = 10 \quad x_2 + 3x_1x_2^2 = 57$ initial guesses $x_1 = 1.5 \quad x_2 = 3.5$ solve (1) for x_1: $x_1 = \frac{10 - x_1^2}{x_2}$ (3) $x_2 = 57 - 3x_1x_2^2$ (4)

iteration 1 from (3) $x_1 = \frac{10 - (1.5)^2}{3.5} = 2.21$ from (4) $x_2 = 57 - 3(2.21)(3.5)^2 = -24.33$ (diverging)

$$x_1 = \sqrt{10 - x_1x_2} \qquad x_2 = \sqrt{\frac{57 - x_2}{3x_1}}$$

Eigenvalues eig(A)

$\underline{\underline{A}}\underline{x} = \lambda \underline{x} \quad \underline{\underline{A}}\underline{x} = \lambda \underline{\underline{I}}\underline{x} \quad (\underline{\underline{A}} - \lambda \underline{\underline{I}})\underline{x} = \underline{0} \quad |\underline{\underline{A}} - \lambda \underline{\underline{I}}| = 0 \quad \lambda_{1,2} = \frac{(a_{11}+a_{22}) \pm \sqrt{(a_{11}+a_{22})^2 - 4a_{12}a_{21}}}{2}$

$(10-\lambda)x_1 - 5x_2 = 0 \quad \lambda_1 = 15$

$-5x_1 + (10-\lambda)x_2 = 0 \quad \lambda_2 = 5 \quad x = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

Gradient tells us what direction to go locally for gaining max elevation.

Gradient

$$\nabla f = \frac{\partial f}{\partial x}\underline{i} + \frac{\partial f}{\partial y}\underline{j} \quad \text{or} \quad \nabla f = \begin{bmatrix} \partial f/\partial x \\ \partial f/\partial y \end{bmatrix}$$

$f(2,2) = 2(2)^2 = 8 \quad \frac{\partial f}{\partial x} = y^2\Big|_{y=2} = 4 \quad \frac{\partial f}{\partial y} = 2xy\Big|_{\substack{x=2 \\ y=2}} = 8$

Gradient $= \begin{bmatrix} 4 \\ 8 \end{bmatrix} \quad \nabla f = 4i + 8j \quad \theta = \tan^{-1}\left[\frac{8}{4}\right] \simeq 63^\circ$

slope in that direction: $\|\nabla f\| = \sqrt{4^2+8^2} = 8.944 \Rightarrow$ we'll gain units of elevation for a unit distance travelled along the steepest path.

The Hessian

$$H = \begin{bmatrix} \partial^2 f/\partial x^2 & \partial^2 f/\partial x \partial y \\ \partial^2 f/\partial y \partial x & \partial^2 f/\partial y^2 \end{bmatrix} \quad |H| = \frac{\partial^2 f}{\partial x^2}\frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2$$

δ: small perturbation

If $|H| > 0$ and $\frac{\partial^2 f}{\partial x^2} > 0 \Rightarrow f(x,y)$ has a local minimum.

If $|H| > 0$ and $\frac{\partial^2 f}{\partial x^2} < 0$ maximum

If $|H| < 0$ then $f(x,y)$ has a saddle point

Finite Diff Approx. Derivatives

$$\frac{\partial f}{\partial x} = \frac{f(x+\delta x, y) - f(x-\delta x, y)}{2\delta x} \qquad \frac{\partial f}{\partial y} = \frac{f(x, y+\delta y) - f(x, y-\delta y)}{2\delta y}$$

maximize $f(x,y) = 2xy + 2x - x^2 - 2y^2$

initial guess $x=-1 \quad y=1$

Analytical solution $\frac{\partial f}{\partial x} = 2y + 2 - 2x = 0$

$\frac{\partial f}{\partial y} = 2x - 4y = 0$ } solve $x=2$, $y=1$ optimum pt.

$\frac{\partial^2 f}{\partial x^2} = -2 \quad \frac{\partial^2 f}{\partial y^2} = -4 \quad \frac{\partial^2 f}{\partial x \partial y} = 2$

$|H| = -2(-4) - 2^2 = 4 > 0$ and

$\frac{\partial^2 f}{\partial x^2} < 0 \rightarrow f(2,1)$ is a max

$f(x,y) = 2xy + 2x - x^2 - 2y^2$ develop 1D version of eqn along gradient at $x=-1 \quad y=1$

$\frac{\partial f}{\partial x} = 2y + 2 - 2x\Big|_{\substack{x=-1 \\ y=1}} = 6 \qquad \frac{\partial f}{\partial y} = 2x - 4y\Big|_{\substack{x=-1 \\ y=1}} = -6$

$x = -1 + 6h \quad y = 1 - 6h$

$f(x,y) = f(-1+6h, 1-6h)$

$g(h) = -180h^2 + 72h - 7$

Polynomial Interpolation

$f(x) = P_1x^2 + P_2x + P_3$

$0.616 = P_1(300)^2 + P_2(300) + P_3$

$0.525 = P_1(400)^2 + P_2(400) + P_3$

$0.457 = P_1(500)^2 + P_2(500) + P_3$

$$\begin{bmatrix} 90000 & 300 & 1 \\ 160000 & 400 & 1 \\ 250000 & 500 & 1 \end{bmatrix}\begin{bmatrix} P_1 \\ P_2 \\ P_3 \end{bmatrix} = \begin{bmatrix} 0.616 \\ 0.525 \\ 0.457 \end{bmatrix}$$

Newton Interpolation Polynomial

Linear interpolation

$$\frac{f_1(x) - f(x_1)}{x - x_1} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

Arrange: $f_1(x) = f(x_1) + \frac{f(x_2) - f(x_1)}{x_2 - x_1}(x - x_1)$

Quadratic Interpolation

$f_2(x) = b_1 + b_2(x - x_1) + b_3(x - x_1)(x - x_2) \quad f(x_1) = b_1 \quad b_2 = \frac{f(x_2) - f(x_1)}{(x_2 - x_1)}$

$$b_3 = \frac{\frac{f(x_3) - f(x_2)}{x_3 - x_2} - \frac{f(x_2) - f(x_1)}{x_2 - x_1}}{x_3 - x_1}$$

$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{f(x_i + \Delta x) - f(x_i)}{\Delta x}$

LU

$LU\begin{bmatrix} x \\ z \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \quad L \cdot \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$

$\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$